4.13.21: Incentre of a Triangle

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Question

The X coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1), (1, 1) and (1, 0) is: (2013)

1. $2+\sqrt{2}$
2. $2-\sqrt{2}$
3. $1+\sqrt{2}$
4. $1-\sqrt{2}$

Step 1: Finding the Vertices from Midpoints

Let the position vectors of the midpoints of sides BC, AC, and AB be $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{F}$ respectively.

$$\mathbf{D} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1}$$

The position vectors of the vertices of the triangle, $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, can be found from the midpoint vectors.

$$\mathbf{A} = \mathbf{E} + \mathbf{F} - \mathbf{D} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \tag{2}$$

$$\mathbf{B} = \mathbf{D} + \mathbf{F} - \mathbf{E} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{3}$$

$$\mathbf{C} = \mathbf{D} + \mathbf{E} - \mathbf{F} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \tag{4}$$

Next, we find the lengths of the sides of $\triangle ABC$. Let a, b, and c be the lengths of the sides opposite to vertices A, B, and C.

Solution

$$a = \|\mathbf{C} - \mathbf{B}\| = \left\| \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\| = \sqrt{0^2 + 2^2} = 2 \tag{5}$$

$$b = \|\mathbf{A} - \mathbf{C}\| = \left\| \begin{pmatrix} 2 \\ -2 \end{pmatrix} \right\| = \sqrt{2^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2} \tag{6}$$

$$c = \|\mathbf{B} - \mathbf{A}\| = \left\| \begin{pmatrix} -2 \\ 0 \end{pmatrix} \right\| = \sqrt{(-2)^2 + 0^2} = 2 \tag{7}$$

The position vector of the incentre, $\mathbf{I}$, is given by the formula:

$$\mathbf{I} = \frac{a\mathbf{A} + b\mathbf{B} + c\mathbf{C}}{a + b + c} \tag{8}$$

Solution

Substituting the vectors and side lengths:

$$\mathbf{I} = \frac{2\begin{pmatrix}2\\0\end{pmatrix} + 2\sqrt{2}\begin{pmatrix}0\\0\end{pmatrix} + 2\begin{pmatrix}0\\2\end{pmatrix}}{2 + 2\sqrt{2} + 2} \tag{9}$$

$$= \frac{1}{4 + 2\sqrt{2}}\left(\begin{pmatrix}4\\0\end{pmatrix} + \begin{pmatrix}0\\0\end{pmatrix} + \begin{pmatrix}0\\4\end{pmatrix}\right) \tag{10}$$

$$= \frac{1}{4 + 2\sqrt{2}}\begin{pmatrix}4\\4\end{pmatrix} = \begin{pmatrix}\frac{4}{4+2\sqrt{2}}\\ \frac{4}{4+2\sqrt{2}}\end{pmatrix} \tag{11}$$

The x-coordinate of the incentre, I_x, is the first component of $\mathbf{I}$.

$$I_x = \frac{4}{4 + 2\sqrt{2}} = \frac{2}{2 + \sqrt{2}} \tag{12}$$

$$= \frac{2(2 - \sqrt{2})}{(2 + \sqrt{2})(2 - \sqrt{2})} = \frac{2(2 - \sqrt{2})}{4 - 2} = 2 - \sqrt{2} \tag{13}$$

C Code - Verification

```c
#include <stdio.h>
#include <math.h>

typedef struct {
    double x, y;
} Point;

Point make_point(double x, double y) {
    Point p; p.x = x; p.y = y; return p;
}

double dist(Point p1, Point p2) {
    return sqrt(pow(p1.x - p2.x,2) + pow(p1.y - p2.y,2));
}

Point find_incentre(Point A, Point B, Point C) {
    double a = dist(B,C);
    double b = dist(A,C);
    double c = dist(A,B);
```

C Code - Verification

```c
    Point I;
    I.x = (a*A.x + b*B.x + c*C.x) / (a+b+c);
    I.y = (a*A.y + b*B.y + c*C.y) / (a+b+c);
    return I;
}

int main() {
    // Vertices derived from midpoints
    Point A = make_point(2,0);
    Point B = make_point(0,0);
    Point C = make_point(0,2);

    Point I = find_incentre(A,B,C);
    printf(Numeric Incentre: (%.4f, %.4f)\n, I.x, I.y);
    printf(Value of 2-sqrt(2) is approx %.4f\n, 2-sqrt(2));
    return 0;
}
```

Python + C Code

```
import ctypes
import matplotlib.pyplot as plt

# Define struct
class Point(ctypes.Structure):
    _fields_ = [(x, ctypes.c_double), (y, ctypes.c_double)]

lib = ctypes.CDLL(./incentre.so)
lib.find_incentre.restype = Point
lib.find_incentre.argtypes = [Point, Point, Point]

# Midpoints
D = Point(0,1)
E = Point(1,1)
F = Point(1,0)
```

Python + C Code

```
# Vertices
A = Point(2,0)
B = Point(0,0)
C = Point(0,2)

# Call C function
I = lib.find_incentre(A,B,C)

# Plot triangle
plt.figure()
plt.plot([A.x,B.x,C.x,A.x], [A.y,B.y,C.y,A.y], 'b-')

points = {
    A(2,0): A,
    B(0,0): B,
    C(0,2): C,
    D(0,1): D,
    E(1,1): E,
    F(1,0): F
```

Python + C Code

```
for (label,pt),col in zip(points.items(),colors):
    plt.scatter(pt.x, pt.y, color=col)
    plt.text(pt.x+0.1, pt.y+0.1, label, fontsize=9)

# Incentre exact + numeric
plt.scatter(I.x, I.y, color='orange')
plt.text(I.x+0.1, I.y+0.1, I(2-2,2-2), color='orange', fontsize=9)

plt.title(Triangle with Midpoints and Incentre (C + Python))
plt.grid(True)
plt.axis(equal)
plt.show()
```

Pure Python Code

```python
import numpy as np
import matplotlib.pyplot as plt

# Midpoints
D = np.array([0,1])
E = np.array([1,1])
F = np.array([1,0])

# Vertices using midpoint relations
A = E + F - D # (2,0)
B = D + F - E # (0,0)
C = D + E - F # (0,2)

# Side lengths
a = np.linalg.norm(C-B)
b = np.linalg.norm(A-C)
c = np.linalg.norm(A-B)
```

Pure Python Code

```
# Incentre (numeric)
I = (a*A + b*B + c*C) / (a+b+c)

# Plot triangle
plt.figure()
plt.plot([A[0],B[0],C[0],A[0]], [A[1],B[1],C[1],A[1]], 'b-')

# Mark points with exact labels
points = {
    A(2,0): A,
    B(0,0): B,
    C(0,2): C,
    D(0,1): D,
    E(1,1): E,
    F(1,0): F
}
colors = ['red','green','blue','purple','brown','cyan']
```

Pure Python Code

```
for (label,pt),col in zip(points.items(),colors):
    plt.scatter(pt[0], pt[1], color=col)
    plt.text(pt[0]+0.1, pt[1]+0.1, label, fontsize=9)

# Incentre label in exact form
plt.scatter(I[0], I[1], color='orange')
plt.text(I[0]+0.1, I[1]+0.1, I(2-2,2-2), color='orange', fontsize
    =9)

plt.title(Triangle with Midpoints and Incentre)
plt.grid(True)
plt.axis(equal)
plt.show()
```

Plot of the Triangle and its Incentre